

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Specification for Pipeline Cleaning, Gauging, & Hydrotesting

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Revision History

Rev No.	Date	Description of Revision
R01	22/07/22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-010

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)
EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Wall Thickness Calculation Report
EA-BFPCL-GPMC-GGPL-FEED-PPL-ALS-001 to 038	Pipeline Alignment Sheet
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009	Pipeline Construction Specification.
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-010	Pipeline Cleaning, Gauging, & Hydrotesting Specification.
EA-BFPCL-GPMC-GGPL-FEED-PPL-PHY-001	Pipeline Integrity Management Philosophy.

Table of Contents

Abbreviations	6
1. INTRODUCTION	7
1.1. Background Information	7
1.2. Purpose	8
2. REGULATIONS, CODES AND STANDARDS	8
3. ENVIROMENTAL AND LINE PIPE DATA	10
3.1. 24" Multiphase Pipeline Data	10
3.2. 36" Multiphase Pipeline Design Data	10
3.3. Environmental Data	11
4. BASIC REQUIREMENTS	12
4.1. Assumed Construction Status	12
4.2. Pre-Commissioning Philosophy	12
5. CONTRACTORS SCOPE OF WORK	14
5.1. Performance Requirements	14
6. DOCUMENTATION	16
6.1. Pre-Test Requirements	16
6.2. Daily Site Requirements	16
6.3. Post Testing Requirements	16
7. MATERIALS AND EQUIPMENT AND INSTRUMENTATION	18
7.1. Materials	18
7.1.1. Water Supply and Quality	18
7.2. Equipment and Instrumentation	19
7.2.1. Pigs	20
7.2.2. Pig Signalling and Pig Location	20
7.2.3. Gauge Plates	20
7.2.4. Baseline Intelligent Pigging	21
8. HYDROSTATIC TEST BASIC REQUIREMENTS	22
9. CLEANING, GAUGING HYDROSTATIC TESTING AND PRE-COMMISSIONING	23
9.1. Basic Requirements	23
9.2. Sequence of Operations	24
9.3. System Leak Testing	25
9.4. Pre-Commissioning	25
9.4.1. Dewatering	25
9.4.2. Lay-Up and Preservation	26
9.4.3. Commissioning	26
10. SAFETY AND ENVIRONMENT	27

10.1. Contingency Plans and Procedures	27
10.2. Hydrocarbon /Chemical Spills	28
10.3. Water Sources	28
10.4. Hydrotest Water Disposal And Waste Management	28
10.5. Company and DPR Approvals	28

Abbreviations

Abbreviation	Meaning
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
CS	Carbon Steel
DEP	SPDC- Design and Engineering Practices
DP	Design Pressure
FEED	Front End Engineering
ISO	International Organisation for Standardization
3LPE	Three-Layer Polyethylene
MMscf/d	Million standard cubic feet per day
NPS	Nominal Pipe Size
OD	Outer Diameter
PE	Polyethylene
PSL	Product Specification Level
DT	Design Temperature
SPDC	Shell Petroleum Development COMPANY Nig. Limited
SMTS	Specified Minimum Tensile Strength
SMYS	Specified Minimum Yield Strength
WT	Wall thickness
WPS	Welding Procedure Specification
WPQR	Welding Procedure Qualification Record

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

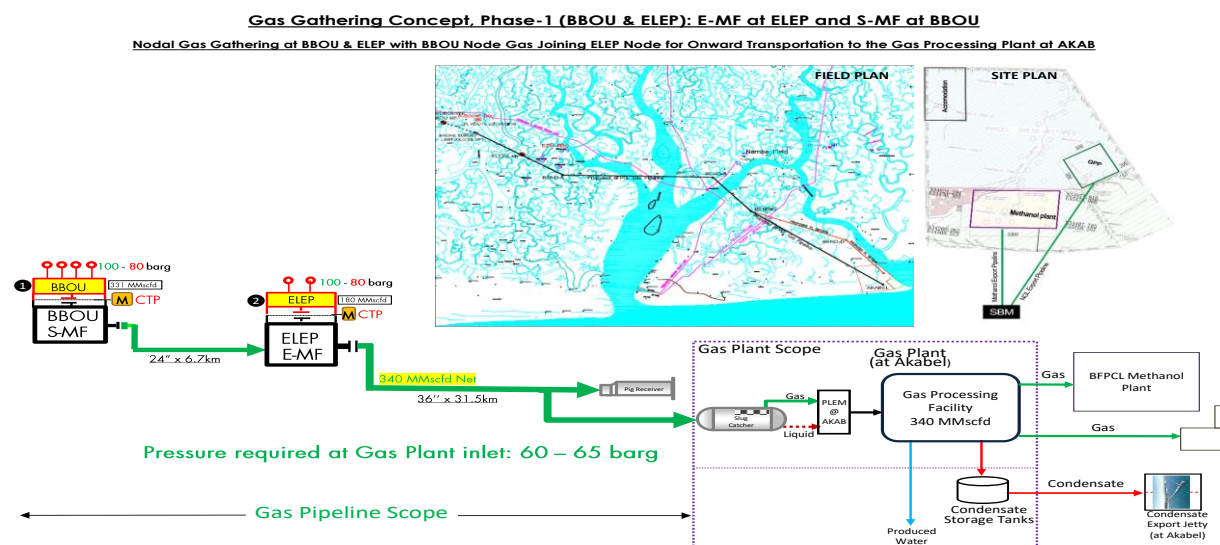


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to present the minimum requirements on Cleaning, Gauging and Hydrostatic Testing, as part of Pre-Commissioning activities on completion of installation of the pipeline systems of BFPCL Gas Gathering Pipelines & Associated Infrastructure Project (Phase1). The activities include but not limited to the following:

- Line Filling
- Hydrostatic Testing
- Dewatering
- Drying and
- Off-plot pipeline sections

The sequence shall be Cleaning, Gauging (including calliper pigging), Filling, Pressure Test, Dewatering and Drying.

The above activities shall be carried out in accordance with DEP 31.40.40.38-Gen and DEP31.40.50.30-Gen taking due regard of Health, Safety and Environmental requirements, including strict compliance to "Life Saving Rules".

The Pipeline system are in two segments as shown in section 3 of this document.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.40.00.10-Gen	Pipeline Engineering
DEP 31.40.40.38-Gen	Hydrostatic Pressure Testing of New Pipelines
DEP 31.40.50.30-Gen	Pre-commissioning of Pipelines
DEP. 61.40.20.30-Gen	Welding of Pipelines & Related Facilities (Amendments / supplements to API Std 1104)
ASME B31.8	Gas Transmission and Distribution Piping Systems
API 5L	Specification for Line Pipe
API RP 1111	Design, Construction, Operation and Maintenance of Offshore Hydrocarbon Pipelines and Risers
ASME B31.8	Gas Transmissions and Distribution Piping Systems
ISO 9001	Quality Management Systems - Requirements
ISO 13623	Petroleum and natural gas industries — Pipeline transportation systems
ISO 4001	Environmental Management System

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Enviromental and Line Pipe Data

3.1. 24" Multiphase Pipeline Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L-X70(PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength (SMYS)	483 Mpa
NPS	24"
Density	7,850 kg/m ³
Young's Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Ultimate Tensile Strength	570 MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
Wall Thickness	15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	60mm
Installation Condition	Trenching
Cathodic Protection	Impressed current

3.2. 36" Multiphase Pipeline Design Data

The 36" pipeline design data for the Multiphase Pipeline Project are as shown below

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Design Life	30 years
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature	-13.3°C
Line Pipe Material Grade	API 5L-X70 (PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	36"
Density	7,850 kg/m ³
Young's Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Ultimate Tensile Strength	570MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
Wall Thickness	23.83mm
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	85mm
Installation Condition	Trenching
Cathodic Protection	Impressed current

3.3. Environmental Data

Refer to the environmental data contained in the document EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 Basis of Design

4. Basic Requirements

The Hydrotest, Cleaning and Gauging are pre-commissioning activities to demonstrate the fitness for purpose of the as-installed pipeline, and to leave the pipeline system in an acceptable condition, prior to commencement of operation. The programme of testing and pre-commissioning activities must take account of other operations in the construction process.

The hydrostatic testing of the pipeline shall be in accordance with ASME B31.8, Ref. [10].

These testing and pre-commissioning activities will entail the following:

- Flooding the pipeline with filtered and treated water.
- Cleaning to remove any solid debris from the pipeline and to prevent damage to valves and instrumentation. (This activity is undertaken as part of the pipeline flooding operation)
- Pipeline gauging to demonstrate that the pipeline is free from dents, buckles or other out-of-roundness features, as per ASME B31.8, Ref. [10]. (This activity is undertaken as part of the pipeline flooding operation).
- Strength testing of the as-installed pipeline;
- Leak testing of the entire pipeline system (pig trap to pig trap)
- De-watering; this must remove virtually all water from the pipeline system. Drying will be required in order to prevent an increase in the corrosion potential, or hydrate formation during start up. The removed water must be disposed in an environmentally acceptable manner, leaving the line in a suitable condition for gas flow service.

4.1. Assumed Construction Status

The methodology outlined in this document is predicated upon the following assumed construction status prior to testing / pre-commissioning operations commencing:

- Unacceptable pipeline spans are corrected prior to pipeline flooding / strength testing operations, as appropriate
- The pipeline sections are installed air filled
- The pipe spools will be strength tested as part of the Factory Acceptance Test
- The whole pipeline system will subsequently be leak tested

4.2. Pre-Commissioning Philosophy

The pipeline will be installed buried, complete with the Pig Launcher and Receiver combined. It will be installed empty. The launcher will have been pre-loaded with the required number and type of pigs. The Pig Launcher and / Receiver will be rated to the full test pressure.

Following pipe lay, the line will be flooded, cleaned and gauged with chemically treated water. On successful completion of the flooding, cleaning and gauging activities, the pipeline will be strength tested. Both of these operations will be carried out as on-site activities.

The pipe spools after manufactured, will be cleaned, gauged and hydrotested prior to leaving the fabrication site.

After installation of the pipeline system, the system leak test will be performed. On completion of the system leak test, de-watering of the pipeline system will take place.

The pipeline system will be de-watered in two segments; from BBOU Manifold Platform to Elepa Manifold Platform for the 24" line and From Elepa Manifold Platform to Akabel Piping station for the 36" line using a dual diameter pigging train, separated by slugs. The pig train will be propelled by Nitrogen.

Methanol shall be made available for de-watering and conditioning of the pipeline system. Samples of slug between pigs will be collected and analysed for water content.

5. Contractors Scope of Work

The testing and pre-commissioning scope for the Pipeline systems shall be from permanent pig launchers to permanent pig receivers.

The work shall include but not limited to:

- Develop a pipeline pressure test and Pre-commissioning procedures in line with this Specification.
- Perform all Engineering works necessary for the completion of these activities.
- Provide all equipment and material requisition and procurement including transportation, delivery to sites, customs clearance, handling, storage and incidental matters relating to temporary works.
- Supply all certificates, qualifications, calculations, reports, drawings and documentation required by Government agencies, inspection institutions and surveyors.
- Carry out all construction activities required for the performance of testing and pre-commissioning works. These works shall include but not limited to the following;
 - cleaning, gauging, flooding, testing, dewatering, dehydration and conditioning of the new pipelines.
- In the event that commissioning of the pipeline systems is delayed, Contractor shall develop a procedure that shall be approved by COMPANY for the preservation of the pipelines.
- Carry out repairs to damaged pipelines caused by failure to properly carry out work.
- Repairs and re-test in case of leaks/failure of test.
- Disposal of water in a safe manner and to a safe location as agreed by COMPANY.

Interface coordination shall be required with other parties in the project. CONTRACTOR in conjunction with COMPANY shall evaluate interface scope during the design, planning, testing and pre-commissioning stages of the Pipelines.

5.1. Performance Requirements

CONTRACTOR shall implement requirements of this document and perform the testing and pre-commissioning works in accordance with the requirements of DEP 31.40.40.38-Gen and DEP 31.40.50.30-Gen.

CONTRACTOR shall ensure that the minimum performance requirements are achieved in a manner that is inherently safe and reliable as demanded by this specification.

CONTRACTOR shall prepare a Quality Plan covering aspect of all works.

CONTRACTOR shall be responsible for the supply and disposal of Hydrotest and all water required for Pre-commissioning in accordance with DEP 31.40.40.38-Gen.

CONTRACTOR shall provide test report detailing the water chemical constituents before commencement of any phase of the Work

6. Documentation

6.1. Pre-Test Requirements

In addition to the documents identified in DEP 31.40.40.38-Gen and DEP 31.40.50.30-Gen, the CONTRACTOR shall as a minimum submit the following documents for review by COMPANY.

- Scope of Work including any approved hydrotest procedures.
- Hydrotest Procedure approved by Shell
- Quality Plan, Inspection and Test Plan (ITP)
- Schedule
- Pipeline weld/NDT traceability records
- Equipment layout drawings
- Air content determination, temperature stabilization calculations
- Equipment certificates
- Proposed hydrotest water chemical analysis report
- Pipeline drawings including PEFS, Pipeline profile and
- Hydrotest equipment and proposed test water chemical analysis
- Description of the test activities and equipment, with diagrams (PEFS and pipeline route profile)
- P&ID with Pipeline profile, indicating different WT section and depth (Indicating lowest & highest point with dimensions)
- Plan for line in-fill water disposal
- All procedures and information demanded in this report
- Gauge plate Inspection & acceptance report
- Temporary Installations for Pressure Test -Inspection Reports

6.2. Daily Site Requirements

- Daily progress reports
- Weather conditions
- Log of activities
- Present progress and program for next 24hrs
- Equipment failures and repairs
- Personnel on Board (POB)
- Any problems encountered.

6.3. Post Testing Requirements

For each completed pipeline, the CONTRACTOR shall, as a minimum and in addition to those required by DEP 31.40.40.38-Gen and DEP 31.40.50.30-Gen, submit the following documents

to COMPANY:

- Safety summary/log.
- Records of all operational parameters requiring company approvals.
- Hydro-test (Strength/leak) measured pressure chart
- Hydrostatic test report.
- Cleaning / gauging reports
- Pigging Records;
- Completed Inspection and Test Plan.
- Original scope of work including any deviations.
- Description of test activities and equipment with diagram.
- Pipeline system profile.
- Instrument calibration certificates.
- Air content determination and temperature Stabilisation Calculations Report of all defects or failures.

7. Materials and Equipment and Instrumentation

All works defined in this document shall be performed in accordance with established industry safe practice, taking due regard of Health Safety and Environmental requirements.

7.1. Materials

All materials required for cleaning, gauging, flooding, hydrostatic pressure testing, de-watering and conditioning of the pipeline systems shall be provided by CONTRACTOR.

CONTRACTOR shall ensure that all equipment and instruments are properly maintained in full working order and are suitable for the intended use.

7.1.1. Water Supply and Quality

The Contractor shall ensure that the water to be used for the various pressure testing operations is of a quality that protects the test section from the ingress of foreign matter, sedimentation and internal metal corrosion.

The flooding water should be taken from a river, aquifer or a potable water system and should be chemically treated with oxygen scavenger and biocide, to mitigate the risks of corrosion or bacterial growth. Depending on the length of time that the water will be within the pipeline, the correct dose rate of chemicals will need to be defined.

The Contractor shall be responsible for obtaining all necessary permits from the owner/local Authority concerning the supply and the disposal of line-fill water.

Hydrotest water and all water required for the pre-commissioning shall be supplied and disposed of in accordance with DEP 31.40.40.38-Gen.

The water used for pressure testing should be selected from a location that will minimise the temperature stabilisation period. The water shall be filtered through 50 micron filters before entering the test section. The arrangement of filters at the test work sites should enable back flushing without disconnecting the pipe work.

The Contractor shall obtain samples from all water sources proposed as line-fill water for analysis by a testing laboratory approved by the Company. The analysis shall show:

- Sulphate concentration
- Fatty acids concentrations, if the total dissolved organic carbon is above 15 ppm
- Ammonium concentration.

To minimise biological fouling and/or bacterial corrosion to an acceptable level in the filled pipeline, the above required analysis shall not exceed the following limits:

- Sulphate concentration 42 ppm
- Fatty acids concentration 14 ppm
- Ammonium concentration 3 ppm

7.2. Equipment and Instrumentation

It will be the CONTRACTOR's responsibility to provide all equipment and instrumentation necessary for the pipeline system Flooding, Cleaning, Gauging, Hydrostatic Strength, Leak Testing and De-Watering of the pipeline system, which shall include as a minimum:

- Break tank and piping, sized to ensure at least five (5) minutes of flow at the required minimum rate
- Flooding pump(s), sized to ensure sufficient flow to achieve the minimum pigging speed and the maximum expected pressure
- Filter skid, sized to be capable of flow rate at maximum requirement for intended operations
- Chemical injection skid, with sufficient number of pumps to cover all types of chemical to be pumped, plus 100% spare and contingency pumps. Its size should be capable of operating at the required injection pressure and at the maximum flow of the flooding system
- Test / control cabin, suitably powered and cleaned with facility to mount pressure gauges, recorders and flow metering displays. The cabin should be set up to allow for all operations to be controlled from this point
- Pressure / Flow metering system
 - To include a full range of pressure and flow measurement and recording equipment
 - Flooding equipment maximum pressure should not be less than 50% and not exceed 90% of the range of instrumentation
 - All expected flow conditions should be within the range of the flow metering system
 - The chemical injection system should include a means of accurately measuring and verifying the injection value in parts per million
 - Hoses and manifolds
 - All instrumentation necessary for monitoring and continuous recording of Pigging and Hydrotest activities

The CONTRACTOR must provide current calibration certificates from a recognised agency for pressure and temperature monitoring instrumentation to the COMPANY before commencement of hydrostatic strength testing operations. All instrumentation must have been calibrated within 30 days prior to mobilisation and must not be used on other projects between calibration and mobilisation.

7.2.1. Pigs

The following types of pigs may typically be used, noting that the pigs may be combined into a 'train' to save both time and cost:

Cleaning Pigs Cleaning pigs shall be high sealant bi-directional polyurethane disc/brush type capable of cleaning construction debris and loose material from the pipeline interior. At least one cleaning pig in each train shall be magnetic in nature to facilitate the removal of magnetic debris. They shall be in a 2 x 4 configuration.

Gauging pigs shall consist of a gauging plate mounted on a high sealant bi-directional polyurethane disc pig. The gauging plate shall be designed in accordance with DEP 31.40.40.38-Gen

Flooding pigs shall be a high sealant bi-directional polyurethane disc type capable of cleaning construction debris and loose material from the pipeline interior.

Dewatering pigs shall be a high sealant bi-directional polyurethane disc type capable of removing water from the pipeline.

Foam pigs shall be of standard design, capable of dewatering and drying the pipeline

It should be noted that the design of the above pigs is very similar and indeed some pigs may be utilised for different pigging operations.

7.2.2. Pig Signalling and Pig Location

A suitable pig signaller should be fitted to signal the launch of each pig and the total number of pigs launched, similarly, a pig signaller should be fitted to the pig receiver to signal receipt of each pig and the total number of pigs received. The Non-intrusive type pig signaller is proposed for this project.

Equipment should also be supplied to locate a stuck or stalled pig within the pipeline.

Pig signalling and location equipment data sheets and specifications are to be submitted to COMPANY for review prior to inclusion in specific project procedures.

The volume of fluid injected behind a pig or pigs should not be used as the sole method for the determination of pig location.

7.2.3. Gauge Plates

Gauging pigs will be bi-directional disc type. The gauging plate will be fitted in front of the third disc. Gauge plates will be machined aluminium, 6-12 mm thick and be removable from the pig. The leading edge of each gauge plate will be chamfered 45° to half the plate thickness.

The gauge plate diameter is to be calculated according the following formula:

Plate diameter = minimum nominal diameter * 0.95

An electronic calliper pig may be used as an alternative to gauge plates.

7.2.4. Baseline Intelligent Pigging

The pipeline system will be designed to allow its integrity to be verified during operational field life.

The first in-service intelligent pigging should be performed within 2 years of pipeline start up, depending on the results of corrosion studies.

Subsequent intelligent pig runs should be performed at a frequency not exceeding 3 years.

The flooding of the pipeline presents an opportunity to ultrasonically inspect it. This method is known to be more accurate than the Magnetic Flux Leakage (MFL) technique normally used when pipeline is in service.

8. Hydrostatic Test Basic Requirements

Pressure tests shall only be made against blind flanges, bleed or vent valves, or test heads. All such items shall be certified for use at the test pressure. Hydrostatic tests shall not be made against closed valves that are part of the permanent facilities.

If, during any stage of pressurisation, or the test hold period, should a leak be present which cannot be isolated by means of a double block and bleed, the pressure shall be reduced to static head pressure before remedial work is undertaken.

It is the CONTRACTOR's responsibility to propose suitable equipment and methods of leak detection and location, either at flange locations or along the main pipeline sections. CONTRACTOR must have resources available at short notice should the strength or leak test indicate the possibility of a leak.

The minimum duration for the hydrostatic pressure test shall be a 4-hour strength test followed by a 24-hour leak tightness test. The assembly testing shall comprise a 4-hour strength test followed by visual examination at the leak tightness test pressure.

The CONTRACTOR must submit a procedure to show when the test reached the calculated or the estimated pressure changes due to temperature fluctuation during strength test period.

The acceptance criterion for the hydrostatic strength test(s) is no unaccountable pressure loss. The stabilisation period will only be considered to be completed on approval by the COMPANY representative.

Depressurisation will be performed in a controlled manner at a rate that does not generate vibrations in associated pipework or at a maximum rate of 2 bar/minute. All displaced liquid shall be discharged in accordance with the project procedures. CONTRACTOR shall complete a Certificate of Hydrostatic Pressure Test for the pipeline system tested. Pressure recorder charts shall not be removed until depressurisation is complete.

9. Cleaning, Gauging Hydrostatic Testing and Pre-Commissioning

9.1. Basic Requirements

Reference should be made to the EA-BFPCL-GPMC-GGPL-FEED-OPS-PHY-001 Operations and Maintenance Philosophy with regards pigging of the pipeline system.

CONTRACTOR shall provide comprehensive technical specifications, work methods and procedures, temporary works, equipment and consumables for pipeline testing and pre-commissioning of individual sections of the pipeline system.

Cleaning, gauging and filling of the Pipeline system shall be performed in accordance with the requirements of DEP 31.40.40.38-Gen Ref. [6], unless specified otherwise.

The test pressure at any point in the Pipelines test section shall be at least equal to $1.25 \times$ Design pressure as calculated in Appendix A of this Specification. These shall be agreed with the Company.

The final pressure test shall preclude the option of any 'golden weld'.

The various stages of hydrostatic testing are as contained in DEP 31.40.40.38-Gen. Ref. [6]. The test medium shall be water.

Pre-commissioning shall be carried out in accordance with DEP 31.40.50.30-Gen Ref. [6] to prevent unacceptable internal corrosion and fluid contamination during commissioning and operation of the pipeline; this can be achieved via Sectional Hydrotesting of pipeline sections.

None of the operations defined in this document shall take place before the following documents have been reviewed and approved:

- Pipeline mechanical completion dossier
- Pre-commissioning & Commissioning Procedures
- Test Packs
- All requests for concessions and deviations and changes to the Asset are properly approved.
- statutory authorities are provided with information, when necessary, e.g. at start-up
- As built drawings
- Pipeline material certificates / records
- Pipeline welding and Non-destructive Testing Traceability records

Pipeline radiograph records if a leak is discovered on an element during a leak test, the test shall be repeated after the leak has been repaired. A record of the same shall be documented. COMPANY reserves the right to terminate any tests where in its opinion there is non-compliance to procedures. Furthermore, COMPANY reserves the right to authorise the repeat of tests where unacceptable deviations have occurred.

The pipeline may be hydrostatically tested in sections, in accordance with the approved Hydrostatic Test Diagrams. Prior to the final pressure test, river crossing strings and on-plot piping shall be individually pressure tested. Pipeline test sections shall include all separately installed pipeline sections within the two segments of the pipeline systems in case they cannot be tested end to end. Test sections shall be overlapped so that the final tie-in between test sections can be made without adding a piece of pipe ('pup piece').

Pressure tests shall only be made against blind flanges, bleed or vent valves, or test heads. All such items shall be certified for use at the test pressure. Hydrostatic tests shall not be made against closed valves that are part of the permanent facilities.

If, during any stage of pressurisation, or the test hold period, should a leak be present which cannot be isolated by means of a double block and bleed, the pressure shall be reduced to static head pressure before remedial work is undertaken.

It is the CONTRACTOR's responsibility to propose suitable equipment and methods of leak detection and location, either at flange locations or along the main pipeline sections. CONTRACTOR must have resources available at short notice should the strength or leak test indicate the possibility of a leak.

The minimum duration for the hydrostatic pressure test shall be a 4-hour strength test followed by a 24-hour leak tightness test. The assembly testing shall comprise a 4-hour strength test followed by visual examination at the leak tightness test pressure.

The CONTRACTOR must submit a procedure to show when the test reached the calculated or the estimated pressure changes due to temperature fluctuation during strength test period.

The acceptance criterion for the hydrostatic strength test(s) is no unaccountable pressure loss. The stabilisation period will only be considered to be completed on approval by the COMPANY representative.

Depressurisation will be performed in a controlled manner at a rate that does not generate vibrations in associated pipework or at a maximum rate of 2 bar/minute. All displaced liquid shall be discharged in accordance with the project procedures. CONTRACTOR shall complete a Certificate of Hydrostatic Pressure Test for the pipeline system tested. Pressure recorder charts shall not be removed until depressurisation is complete.

9.2. Sequence of Operations

Hydrostatic pressure testing and pre-commissioning of the pipelines shall be carried out in this sequence. However, improvements to the sequence may be proposed where a cost benefit to COMPANY can be demonstrated:

- Selection of test sections.
- Preparation of the test section through cleaning, gauging and filling.
- Preparation of the hydrostatic test.
- Hydrostatic test to ascertain strength and leak tightness
- Post testing activities through depressurisation and documentation.
- Rectification activities (if required) through leak location during test, dewatering for rectification and rectification of defects.
- Retest line after defect rectification.

9.3. System Leak Testing

A leak test is required to prove the leak tightness of the completed pipeline system. The calculation of the system test pressure must account for all components within the entire as-installed/connected system and specific code requirements / specification breaks relating to each component in the systems.

All components in the system will be verified as being capable of withstanding the maximum calculated system test pressure.

The pipeline should be pre-flooded before installation, or alternatively may be allowed to free-flood on installation, providing that flooding is performed in a manner that prevents sand or other debris from entering the pipework. In either case, the spools must have sufficient inhibitor and ultraviolet-fluorescent dye, which may be in stick or gel form, added prior to tie-in. All chemicals added must be compatible with the chemical treatment.

All pipeline system valves present in the pipeline during leak testing must be fully or partially open and this must be verified prior to commencement of pressurisation. The valves must also have a facility for pressure equalisation across the valve seats, or alternatively, the valve body cavity must be filled with an inert liquid.

9.4. Pre-Commissioning

Pre-commissioning operations shall be performed in accordance with the requirements of DEP 31.40.50.30-Gen. Ref. [6] unless specified otherwise.

9.4.1. Dewatering

Dewatering of the pipeline systems shall be done with due regard to the provisions of the Project Environmental Statement and ISO 14001 Ref. [13], requirements. Procedures for disposal or re-use of the test water for consecutive pipeline tests shall be submitted to COMPANY for approval. Re-use of test water shall be by COMPANY approval and recycled water shall be filtered to the same condition prior to introduction to the previous pipeline section. The recovered solids shall be disposed of in an approved disposal area.

All solids, sludge or other pollutants filtered from the discharged water shall be disposed of in an approved disposal area.

The dewatering of the pipeline will be carried out after successful completion of the system leak testing.

The objective of the de-watering activity is to remove the water prior to commissioning.

It is the CONTRACTOR's responsibility to prepare and submit for COMPANY's approval, complete procedures for de-watering and drying of the pipelines.

9.4.2. Lay-Up and Preservation

Once drying has been completed to a specification acceptable to COMPANY, ahead of commissioning, the pipeline shall be preserved, if required in accordance with DEP 31.40.50.30-Gen. Ref. [6].

9.4.3. Commissioning

It is recommended that commissioning of the pipelines follow the drying and pre-commissioning as closely as possible to maintain planned schedules.

If there are delays in commissioning an assessment to check the condition of dry air package or chemically inhibited water as approved by COMPANY should be undertaken.

10. Safety and Environment

The safety of personnel and the public during the operations is of paramount importance.

Therefore, all safety precautions shall be described in the procedures and must be approved by COMPANY.

Testing and pre-commissioning works shall be carried out in accordance with DEP 31.40.40.38 with due consideration to the constraints imposed by the locations and in full compliance with COMPANY requirements.

These shall include, but not limited to the following:

- Agreed transport routes shall be established.
- Road access shall be strictly controlled.
- Parking areas shall be identified prior to commencement of the work.
- Appropriate measures shall be taken for the disposal of the testing medium and associated waste.
- Compliance with COMPANY requirements in respect of equipment specifications and personnel training.
- Compliance with COMPANY permit-to-work systems
- Precautions to prevent damage to installed pipeline systems, cables and other existing facilities while performing the pre-commissioning activities.
- CONTRACTOR shall be required to develop an HSE plan that demonstrates that

risks associated with pre-commissioning activities have been evaluated and are as low as reasonably practicable.

The main safety requirement as stipulated in the 31.40.40.38-Gen is that CONTRACTOR shall take all necessary measures to ensure that all persons and property are adequately protected from the consequences of a possible system failure during the testing operations. For the period from the start of pressurisation until the reduction in pressure at the end of the test, the Contractor SHALL [PS] ensure that all access to, or work on or near a pipeline section under test is properly authorised taking into account the hazards associated with pressure testing operations.

10.1. Contingency Plans and Procedures

Contractor shall develop procedures and contingency plans to handle potential occurrences which are not in the base plan. This plan and procedure shall be approved by COMPANY. These

occurrences include but not limited to the following:

- Stuck pig.
- Pipeline flow in opposite direction to that covered by the base plan.
- Disposal of chemically treated water from opposite end of pipeline to that covered by the base plan.
- Pipeline repair.

10.2. Hydrocarbon /Chemical Spills

All procedures relating to the handling of hydrocarbons and/or chemicals shall be designed and used to ensure that spills do not occur. Any spills that occur must be reported immediately to COMPANY. Clean up of the sites shall be carried out by the CONTRACTOR in accordance with COMPANY oil spill/chemical spill contingency plans and procedures.

10.3. Water Sources

Only filtered clean fresh water taken from a river, aquifer or a potable water system should be used. Brackish water from estuaries or harbours shall not be used as a source of line-fill water. CONTRACTOR shall be responsible for obtaining all necessary permits from the owner/local authority concerning the supply of line-fill water subject to COMPANY approval.

The line fill-in Water Analysis and compliance shall be in line with DEP 31.40.40.38. Gen

10.4. Hydrotest Water Disposal And Waste Management

Hydro-test water, after use, shall first be held in an evaporation pond to ensure any debris collected from pipe internals are separated and the water subjected to quality test.

All discharge points shall be registered with DPR. All discharges from construction sites shall be monitored and if necessary treated to comply with regulatory limits before disposal.

A documented procedure is required for this, to be approved by the Principal and DPR as necessary.

CONTRACTOR shall assist COMPANY in obtaining all necessary permits from the owner/local authority concerning the disposal of line-fill water subject to DPR approvals.

10.5. Company and DPR Approvals

All technical specifications, work methods, procedures, calculations, drawings identified with this document shall be approved by COMPANY and DPR as required. COMPANY approvals shall not relieve CONTRACTOR of own responsibilities.

All works MUST be performed in the presence of COMPANY representatives. Prior written approvals shall be obtained from COMPANY for all deviations from procedures.